Towards Bayesian Joint Modelling of Ranks and Scores in the Assessment of Grant Proposal Quality

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by

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Abstract

Introduction: Funding agencies rely on expert evaluations to assess the quality of research proposals submitted to funding calls. Most often, experts are asked to score proposals on an ordinal or continuous scale. Some funders instead ask experts to rank the proposals from best to worst, and few funders ask experts to both score and rank proposals. The scoring and ranking data created in this process are then analysed in order to take funding decisions. While many statistical models exist to analyse these data, there curently exists no formal comparison of these models. Moreover, while there already exist a frequentist model for jointly analysing scoring and ranking data, a Bayesian approach to joint modelling is yet to be explored.

Methods: This work formalises the concepts of ranking and scoring data in the context of grant proposal quality assessment and funding decision making. We review and compare six different frequentist and Bayesian methods that can be used for such data. We apply them on a real-world dataset containing both scoring and ranking data collected by the *American Institute for Biological Sciences*. We also present and compare two different methods for posterior distribution estimation in the Bayesian models: the Integrated Nested Laplace Approximation (Inla), and a Markov chain Monte Carlo simulation (Mcmc).

Results: In the data analysis, each of the six models used provides a different ordering of the grant proposals with respect to their quality. The choice of model thus has a direct impact on the estimation of proposal quality, and subsequently can have a direct impact on the funding decision. The choice of Bayesian posterior distribution estimation method did not create any difference in the results. However, the Inla implementation is considerably easier to use than the Mcmc simulation due to it requiring fewer convergence diagnostics.

Discussion: The application of these models on one dataset enables us to compare the results provided by each model individually. We show and discuss that some methods for ranking data favour proposals that consistently obtain middling ranks, while others favour proposals which polarise and obtain both very high and very low ranks. This demonstrates that the choice of model used to analyse the grant proposal assessment data has a direct impact on funding decisions and highlights the importance of making conscious and intentional choices in this context. While these findings are currently limited to the dataset analysed, further work could replicate the analyses performed here on other real-world and simulation data and provide robust model recommendations. Based on the results, we also propose possible approaches for combining scoring and ranking data in Bayesian joint models. In addition, we provide a minimal implementation of such a method using the INLA framework. Altogether, this thesis provides a first comparison of models used for analysing grant proposal data, and provides the foundations necessary for further research in this domain.

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1 Introduction

1.1 Grant Proposal Funding

When resources are limited, decisions have to be made on how to allocate them. In research, funding agencies make this problem by deciding whether to accept or reject grant proposals. Agencies contact panels of experts which individually review, rank or score proposals, the best of which are funded and the rest rejected (Harman 1998). However, there is evidence for bias in this process. For instance, funding for innovative, risky or interdisciplinary research is less likely to be awarded than conservative and conventional research (Guthrie et al. 2019). Moreover, evaluations from individual experts are known to be unreliable, even after panel meetings, making it hard to objectively assess which proposals are worth funding (Fogelholm et al. 2012). Despite these shortcomings, there is no existing alternative to this expert peer review system. To mitigate its downsides, statistical methods can be used to analyse the experts' evaluations in order to ensure fair decision making.

Depending on the assessment system, the experts either make qualitative or quantitative statements about the quality of the proposals. This project will focus on statistical methods used in the data analysis of two quantitative expert evaluation systems:

1. Scoring: Where experts give grades to proposal on either a continuous or ordinal scale.
2. Ranking: Where experts rank all, or a subset, of the proposals from best to worst.

To evaluate grant proposal quality from these data, multiple statistical models, both frequentist and Bayesian, have been developed. These methods have yet to be formally compared in the context of grant proposal quality estimation. In this thesis, we propose a unified notation framework to write about these models. Using this framework, we review and reproduce six statistical methods for evaluating scoring data and ranking data separately and jointly. We apply them to a real-world dataset, and compare the results provided by each model. Finally, we provide paths that can be taken towards creating novel joint models combining scoring and ranking data to provide quality estimates.

1.2 Definitions

Many different notations and formulations are used in the literature to refer to ranking and scoring data. For clarity, this section defines the notation that we will use in this thesis. It is inspired by Heyard et al. (2022) and Xia (2019).

Objects and Assessors

We define objects as the set of items that are ranked and scored by the assessors. Conversely, we define assessors as the people giving scores and rankings to the objects. In this thesis, the objects will be grant proposals and the assessors will be experts on an evaluation panel.

Definitions: Objects and Assessors

Objects are a set of $n \epsilon \mathbb{N}$ items to be ranked or scored by assessors. In this thesis we will denote the set of objects as $\mathcal{O} = \{o_1, ..., o_n\}$, with a subscript $i \epsilon \{1, ..., n\}$.
Assessors are a set of $m \in \mathbb{N}$ experts who give scores and ranks to the objects $\mathcal{O}$. We will denote the individual assessors with a subscript $j \epsilon \{1, ..., m\}$.

Scores

Scores are obtained by having assessors assign a grade to each object by estimating its quality. Grades are measured on continuous or ordinal scales of different lengths. When scoring, assessors examine each object on an absolute scale; they assess objects individually and compare them to their personal representation of an ideal object. The assessment of each object is assumed to be independent from all other objects assessed (see Figure 1, left). Their assessment is said to be on an absolute scale. In this thesis, we work with continuous scoring data and we use the convention that **low scores represent high quality and high scores represent low quality**.

Definitions: Scores

Scores S_j are a set of grades given to all of $\mathcal{O}$ by an assessor j. The score given to a specific object $o_i \epsilon \mathcal{O}$ by a specific assessor j will be written y_{ij} with $i \epsilon \{1, ..., n\}$ and $j \epsilon \{1, ..., m\}$.
Scoring data $S = \{S_1, ..., S_m\}$ are the observed scores given by all assessors.

Rankings

In contrast to scores, rankings are obtained by having each assessor directly compare all objects against each other and order them from best to worst. The objects are therefore assessed on a comparative scale; the assessment of each object is directly influenced by the quality of all other objects in the set of objects evaluated (see Figure 1, right). In this thesis, we will use the convention that **the best rank is the 1st, and then, as the rank ascends, the quality of the objects decreases**.

Definitions: Rankings

Linear orders $\mathcal{L}(\mathcal{O})$ are the set of all possible ranking orders of $\mathcal{O}$. For any pair of objects o_i, $o_l \in \mathcal{O}$, we will denote $o_i \succ o_l$ if o_i is preferred to o_l.
Ranking order $R_j \epsilon \mathcal{L}(\mathcal{O})$ is the linear order given by a specific assessors $j \epsilon \{1, ..., m\}$ to $\mathcal{O}$. The rank position given to an object $o_i \in \mathcal{O}$ by an assessor j will be denoted r_{ij}.
Ranking data $R = \{R_1, ..., R_m\}$ are the observed ranking orders given by all assessors.

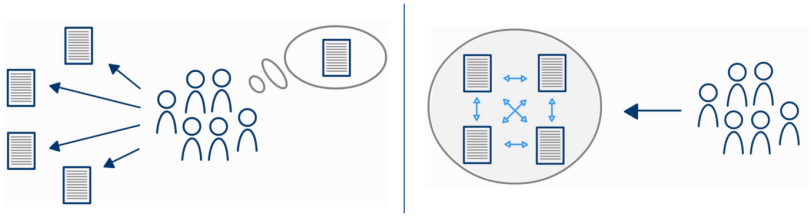


Figure 1: Visualisation of ranking and scoring methodologies. **On the left,** each assessor individually compares all objects to their representation of an ideal object (which remains unobservable) to give scores. **On the right,** each assessor individually compares all objects to each other in order to obtain a ranking.

Object Quality and Final Ordering

The ultimate objective of our modelling will be to select the most appropriate subset of objects for funding, given an available funding amount. To obtain this subset, we make a final ordering of the objects, sorting the objects from best to worst. This is either achieved, in models estimating a latent quality variable, by sorting the latent quality variable in ascending order (lower values of the latent quality variables represent higher object qualities), or, in models which directly calculate a ranking, by taking a model's predicted underlying ranking as the final ordering. Once the final ordering is calculated, a funding line can be introduced. The funding line represents the highest value of the latent quality (the minimal quality standard) necessary for an object to be funded, or, in the case where only a final ordering is available, the limit rank an object needs to achieve in order to be funded.

Definitions: Quality and Final Ordering

Object latent quality θ_i^k is the latent quality of an object o_i according to a model k.
Final ordering $\mathcal{F}_k(\mathcal{O}) \in \mathcal{L}(\mathcal{O})$ is the final ordering of the objects $\mathcal{O}$ according to a model k.
Funding line In the cases where θ_i^k is available, the highest value of the latent quality θ_i^k possible for an object to be funded. In the cases where only $\mathcal{F}_k(\mathcal{O})$ is available, the limit rank an object needs to reach in order for it to be funded.

1.3 Modelling Approaches to Scoring Data and Ranking Data

Models for scoring data are very abundant in the literature. In general, it is assumed that the scores given to objects follow a given distribution from which parameters can be inferred. By interpreting these parameters, statements can be made about the scoring data (for example, regarding the spread of the scores or the expected values of the scores). As ranking data are in a special format, they require extra consideration when being analysed. Consequently, many different formulations and approaches exist to model them. In this section, we will discuss notions of how ranking data can be thought of, described and modelled.

Pairwise Comparisons and Ranking

Ranking data can be thought of as a series of pairwise comparisons. This approach considers that the assessors simply compare many pairs of objects and generate a ranking from these comparisons (Marden 2014). The ranking data are then modelled through a series of parameters representing the probability of winning these pairwise comparisons. This description of ranking data is used in models such as the Bradley-Terry model (Bradley and Terry 1952) and the Plackett-Luce model (Plackett 1975). This description of ranking data comes at a cost as the number of parameters increases rapidly for larger set of objects to be ranked.

Distance-Based Ranking

In distance-based ranking, we assume that the data have a true underlying ranking, and that the assessors are imperfect judges who attempt to reproduce this ranking (Fligner and Verducci 1986). The true ranking is perceived as a ranking order $R^* \epsilon \mathcal{L}(\mathcal{O})$, and the further the observed ranking is from this true ranking, the less likely the ranking is to be present in the data. There are many possible parametrisations of the distance. Using a distance-based model instead of a pairwise comparison model drastically reduces the amount of parameters to estimate when modelling.

Thurstonian Ranking

Thurstonian ranking is another way to model ranking data. As in distance-based ranking, we assume that the assessors are imperfect judges. However, in opposition to distance-based ranking, the ranking data are not assumed to be generated based on a true underlying linear order. Instead, we assume the ranking data to be generated by a series of latent random variables representing each object's quality (Thurstone 1927). According to Thurstonian ranking, each assessor observes a realisation of the latent random variables of all objects and ranks them from highest to lowest based on these observed values (see Figure 2).

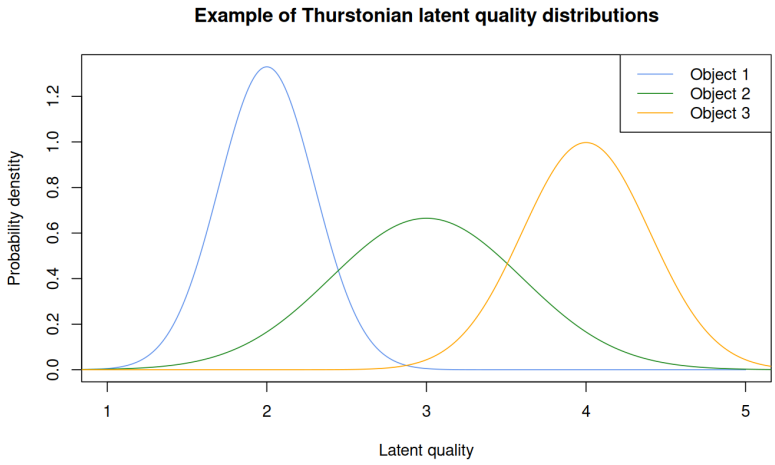


Figure 2: Example of Thurstonian ranking with three objects. The lower the score, the higher the quality. Each assessor observes realisations of the three objects' latent quality distributions. The order of the realisations determines the order in which the assessors rank the objects. In this case, the most common ranking observed will be $o_1 \succ o_2 \succ o_3$. The rankings $o_2 \succ o_1 \succ o_3$ and $o_1 \succ o_3 \succ o_2$ will also commonly be observed, but all other linear orders will very rarely be observed.

2 Data

To compare and reproduce models, we will fit them on a dataset made available in the supplementary material of Gallo et al. (2023). The data were collected in the context of a real-world call for proposals by the *American Institute of Biological Science* (AIBS) in 2020. This dataset contains both scoring data and partial ranking data given to 28 objects (grant proposals) by 10 assessors (experts). The partial ranking is provided for the six best objects according to each assessor and the objects are scored on a scale from 1 (best) to 5 (worst). The funding was awarded to the best 3 objects. A summary of the main properties of the data can be found in Table 1.

Score property	**Value**	**Rank Property**	**Value**
Scoring scale	1 to 5	Ranking scale	Rop 6
Objects (n)	28	Objects (n)	11
Assessors (m)	10	Assessors (m)	10
Scores per object	7-10	Ranks per object	1-10
Missing scores	21	Missing ranks	5
Total scores (obs./tot.)	259/280	Total ranks (obs./total)	49/60

Table 1: Overview of the main characteristics of the dataset from Gallo et al. (2022).

The data are unfortunately incomplete. In the scoring data there are 21 missing observations out of the 280 possible assessor-object scoring observations. Five of these observations are missing due to conflicts of interest (MCI) between the assessors and the object, and 16 of them are assumed to be missing completely at random (MCAR). In the ranking data, 11 observations are missing, five MCI and six MCAR. Specific considerations taken to work around this missingness are discussed in the scoring and ranking data sections below.

2.1 Scoring Data

The inter-assessor reliability in this dataset was calculated using a two-way, mixed effect, fixed rater ICC. We use the 2-way mixed effect version as as all raters are assessing the same set of objects, and we use the fixed rater version as we want to measure the agreement within the fixed individual assessors in this group. The ICC estimate is 0.91 (95% CI from 0.87 to 0.96), which signifies that the grades given by the assessors have good to excellent agreement (Koo and Li 2016). In the exploratory analysis, we found a significant digit preference for the score 3, and no score worse than (i.e. higher than) 4 was given to any of the objects being evaluated (see Figure 3, left). All assessors used approximately the same span of grades (see Figure 3, right). Nevertheless, we believe this digit preference and the lack of high scores will not affect our analysis.

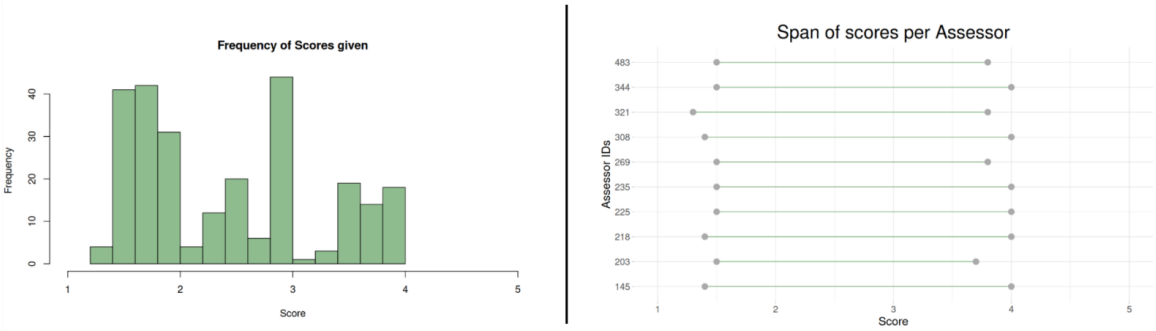


Figure 3: Exploratory analysis of the scoring data: **On the left**, a histogram representing the frequency each score was given. **On the right**, the span of scores given by each assessor.

There are 21 missing data points in the scoring data. This represents 7.5 % of the data, with no single object having more than three missing scores. The MCAR data, despite increasing the variability in object quality estimation, do not create any systematic bias (Dey et al. 2026). The methods we use on scoring data in this thesis have no problems handling missing values and there is therefore little interest in imputating MCAR datapoints. On the other hand, the MCI data are intentionally not collected from the assessor because they are known to be biased. Accordingly, imputating these values is also undesirable. We therefore proceed in our scoring models with a complete case analyses.

2.2 Ranking Data

An overview of the ranking data can be found in Figure 4. Each assessor reported a partial ranking of the six best objects according to their viewpoint. Thus, we obtain information about the ranking on seven levels: ranks one through six, and a rank greater or equal to seven. Assessors were not allowed to consider ties in the positions one to six. The inter-assessor reliability calculated using a two-way, mixed effect, fixed rater ICC is 0.55 (95% CI from 0.33 to 0.8), which signifies that the ranks given by the assessors have a moderate level of agreement (Koo and Li 2016).

There are 11 missing data points in the ranking data, five MCI, and six MCAR, which represent 18.3% of the ranking data. The MCAR data are due to an assessor ignoring the ranking task. We remove them from the dataset as they do not provide any information about the object qualities. However, due to the nature of the models we are using, the MCI data are problematic. The models assume that when an object has a missing value, it did not make it into the best six of an assessor. Reformulated, a missing value given by a specific assessor represents an object quality worse than the 6th ranked object according to them. However, in the case of MCI data, the missing values do not provide any information about the object quality. Therefore, there is a need for data imputation.

Upon further inspection of the data, three of the five MCI datapoints in this dataset represent objects that were ranked in position seven or greater by all assessors. Moreover, the two last MCI datapoints, were each only included in one of the nine assessor's best six objects and given the sixth rank. Given this information, we consider that the MCI datapoints represent object which would have been of worse quality than the assessors best six's, which removes the need for imputation. This can only be assumed due to the nature of this dataset, and would have to be done differently on other datasets.

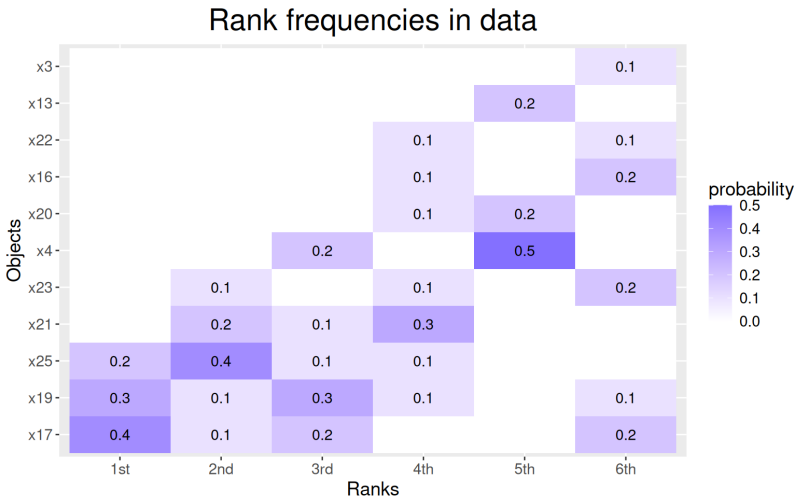


Figure 4: Exploratory analysis of the ranking data: The probabilities represent the frequency that a given object was ranked in a given position.

2.3 Comparisons between Scoring and Ranking Data

All objects that made it into the best 6 objects of at least one assessor only obtained scores $y_{ij} \leq 2$. Eight out of the nine assessors provide a ranking order R_j that is different from sorting the scores y_{ij} they gave to each object in ascending order. Moreover, giving rankings to objects forced all assessors to break ties they gave in the scoring. In order to visualise the relationship between scores and ranks, we plot the mean score and the mean rank of all objects having made it into the ranking of at least one assessor in Figure 5.

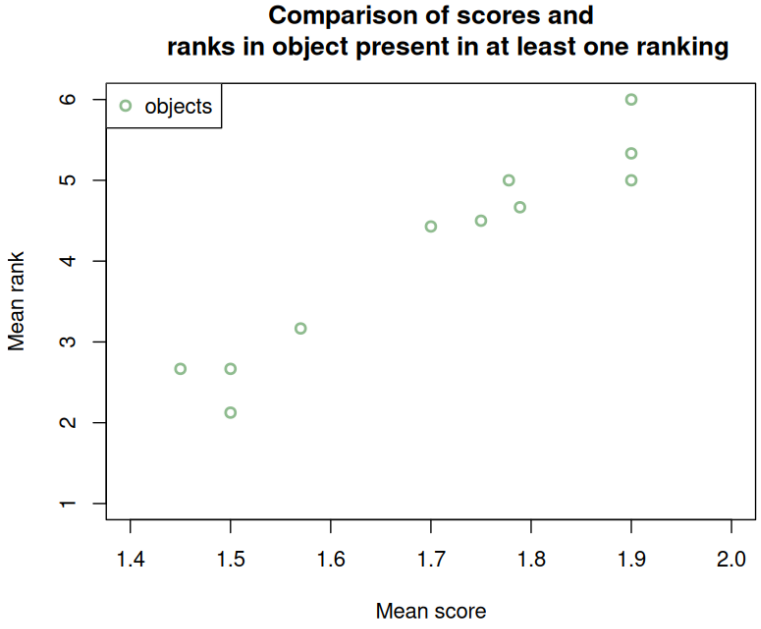


Figure 5: A plot of the mean scores and mean ranks obtained by objects present in at least one of the assessors' rankings: On the x-axis, the mean scores given to the objects calculated from the scores given by all assessor. On the y axis, the mean rank given to each objects calculated from the rankings given to the objects by the assessors which included the object in their best six ranking.

3 Model Descripion & Examples

In this section, we will present a total of six methods that can be used to generate a final ordering from the data described in Section 2. Two of these methods are focused on analysing scoring data, three of them use only ranking data and one jointly models scoring and ranking data. We introduce each method separately, and show the results provided by applying each method to the data. Due to the diversity of models used, we also obtain diverse types of results. In four models we obtain the estimated latent quality $\hat{\theta}_i^k$ of each object, while in two models we obtain the final ordering $\hat{\mathcal{F}}_k(\mathcal{O})$ of all the objects and respectively a rank $\hat{r}_i^k$ for each object. Each model was fitted on the entirety of the dataset. However, for readability, we only provide the results for the ten best objects in each final ordering. An overview of

the methods analysed and their main characteristics can be found in Table 2.

	Scores	**Ranks**	**Frequentist**	**Bayesian**	$\hat{\theta}_i^k$	$\hat{r}_i^k$
Mean scores	✓		✓		✓	
Hierarchical model for scores	✓			✓	✓	
Plackett-Luce		✓	✓		✓	
Mallows		✓	✓			✓
Thurstonian		✓	✓		✓	
Mallows-Binomial	✓	✓	✓			✓

Table 2: Overview of scoring data and ranking data models analysed in this thesis with their main characteristics.

3.1 Models using Scoring Data

Mean Scores

A first and straightforward approach proposed by Zehlike et al. (2022) to analyse the quality of each object is to use the mean of all the scores y_{ij} obtained by an object o_i as an approximation of the object's quality. In this case, the latent quality of an object ${\theta_i}^{mean}$ is estimated as follows:

$$\bar{y}_i = \frac{\sum_{j=1}^{m} y_{ij}}{m}$$
$$\bar{y} = \frac{\sum_{i=1}^{n} \bar{y}_i}{n}$$
$$\hat{\theta}_i{}^{mean} = \bar{y}_i - \bar{y}$$

Note that, by convention, high quality objects will have negative values of latent quality $\hat{\theta}_i{}^{mean}$. This model relies on many strong assumptions. First, it assumes that each assessor has the exact same scoring habits, and is well calibrated with themselves and the rest of the group (for example, that no assessor is stricter or more lenient than their peers). Second, it also assumes that each object is evaluated completely independently from all other objects. In this thesis, we fit this model using an `R` script (R Core Team 2026).

The mean scores model provides an estimate of the latent quality $\hat{\theta}_i{}^{mean}$ of the objects from the scoring data. Figure 6 visualises the latent quality estimates for the ten best objects according to this model. The mean score model would accept objects x17, x19, and x25. Object x19 and x25 are tied for second place, and the difference in latent quality between the tied second place and the fourth place object is very low. The latent quality estimates for the ten best objects according to this model are provided in Table 3.

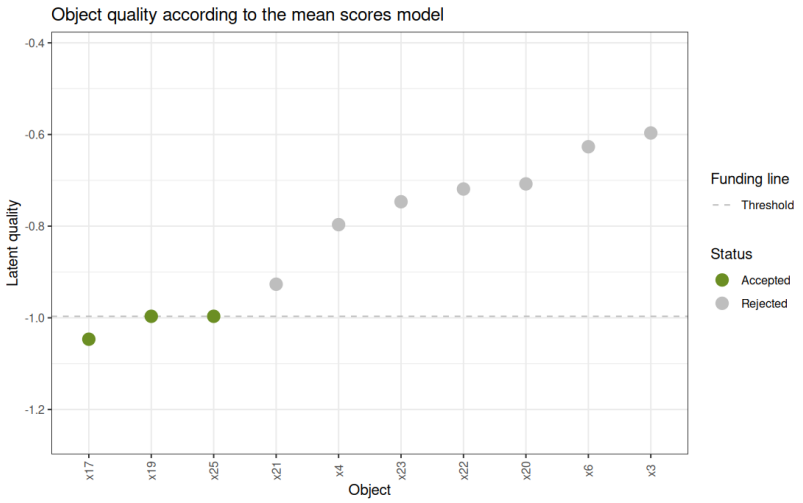


Figure 6: Latent quality estimates of the ten best objects according to the mean scores model (lower values represent higher object qualities).

Grant ID	x17	x19	x25	x21	x4	x23	x22	x20	x6	x3
$\hat{\theta_i}^{mean}$	-1.05	-1.00	-1.00	-0.93	-0.80	-0.75	-0.72	-0.71	-0.63	-0.60

Table 3: Latent quality estimates of the 10 best objects according to the the mean scores model sorted from best to worst.

Hierarchical Model for Scores

To alleviate the assessor calibration assumptions of the mean scores model, Heyard et al. (2022) proposed a hierarchical model for scores that includes parameters to model the different scoring behaviors of the assessors. We define a slightly modified version of this model, with gamma priors instead of flat priors. This modification allows us to fit the model using two different posterior distribution estimation procedure and compare their results. We define it as follows:

$$y_{ij}|{\theta_i}^{hier}, \lambda_j \sim N(\mu + {\theta_i}^{hier} + \lambda_j \ , \frac{1}{\tau_y})$$

$${\theta_i}^{hier} \sim N(0, \frac{1}{\tau_\theta}) \qquad \lambda_j \sim N(0, \frac{1}{\tau_\lambda})$$

with priors:

$$\mu \sim N(0, 1000) \qquad \tau_y \sim G(1, 0.1), \qquad \tau_\theta \sim G(1, 1) \qquad \tau_\lambda \sim G(1, 1)$$

with μ representing and estimating the mean score of all objects, and λ_j being the assessor-specific effect representing the differences in scoring habits and strictness. Given the scores are

grades given between one and five, we fix gamma priors for the precision τ with distributions which lie within plausible values for these parameters, with modes of 1, and standard deviations of 1. Using this model, we obtain and extract the posterior distribution for the latent quality of each object ${\theta_i}^{hier}$, which we use to provide credible intervals of ${\theta_i}^{hier}$. This model can be used in combination with a funding line, Heyard et al. (2022) propose that all proposals that have a 50% credible interval of ${\theta_i}^{hier}$ crossing the funding line should be considered in a lottery instead of being directly accepted or rejected.

This model can be fitted using the `ERforResearch` package (Heyard 2026). This implementation uses the `rjags` (Plummer 2023) framework in order to generate a MCMC sample of the posterior distributions of ${\theta_i}^{hier}$. To fit the model, we will use both a reimplementation of this model in `rjags` with the new priors, and a novel implementation of this model using the Integrated Nested Laplace Approximation INLA framework using the `inla` package (Rue et al. 2009). More information on both these methods can be found in Annex A.3 and Annex A.4. The `rjags` model was called with 10000 iterations of adaptive sampling, 10000 iteration of burn-in and 80000 iterations of sampling in combination with a thinning of 20. Convergence diagnostics of the `rjags` model are available in Annex A.7 and the model calls are available in Annex A.8.

Both fitting methods provide very similar posterior distribution for θ_i^{hier}. Figure 7 visualises the latent quality estimates for the ten best objects according to both the `rjags` fit and the `inla` fit. Both fits would suggest a lottery to allocate the funding to three of the four best objects: x17, x25, x19 and x21. The latent quality estimates for the ten best objects according to both these fits are provided in Table 4. Both model fits provide the same output; the largest discrepancy between the two fitting procedure is of a magnitude of 0.01. However, the `inla` model does not require as extensive convergence diagnostic as the `rjags` model and is thus easier to use.

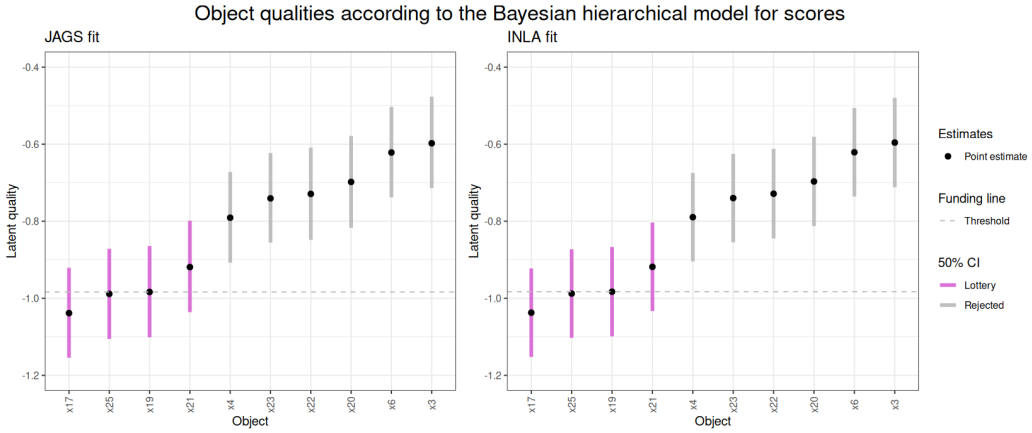


Figure 7: Latent quality estimates of the 10 best objects according to the Bayesian hierarchical model for scores (lower values represent higher object qualities). Credible intervals (CI) of the estimated object qualities are available in purple and grey. **On the left:** The model fitted using `rjags`. **On the right:** The model fitted using `inla`. Both posterior distribution estimation methods yield very similar results.

Grant ID	x17	x25	x19	x21	x4
$\hat{\theta}_i^{mean}$ (`rjags`)	-1.04	-0.99	-0.98	-0.92	-0.79
$\hat{\theta}_i^{mean}$ (INLA)	-1.04	-0.99	-0.98	-0.92	-0.79
50% CI (`rjags`)	-1.15 to -0.92	-1.11 to -0.87	-1.10 to -0.86	-1.04 to -0.80	-0.91 to 0.67
50% CI (INLA)	-1.15 to -0.92	-1.10 to -0.87	-1.10 to -0.87	-1.03 to -0.80	-0.90 to -0.67
Grant ID	x23	x22	x20	x6	x3
$\hat{\theta}_i^{mean}$ (`rjags`)	-0.74	-0.73	-0.70	-0.62	-0.60
$\hat{\theta}_i^{mean}$ (INLA)	-0.74	-0.73	-0.70	-0.62	-0.60
50% CI (`rjags`)	-0.86 to -0.62	-0.85 to -0.61	-0.82 to -0.58	-0.74 to -0.50	-0.71 to -0.48
50% CI (INLA)	-0.85 to -0.63	-0.84 to -0.61	-0.81 to -0.58	-0.74 to -0.51	-0.71 to -0.48

Table 4: Underlying estimated posterior means of the quality estimates and 50% credible intervals (CI) of the 10 best objects according to the hierarchical model for scores fitted with `rjags` and `inla`. Objects are sorted from best to worst according to the hierarchical model for scores.

3.2 Models using Ranking Data

Plackett-Luce

The Plackett-Luce model is a model based on pairwise comparison. It assumes that each object $o_i \in \mathcal{O}$ has a fixed probability of coming in first place when compared with the other objects in the set $\mathcal{O}' = \mathcal{O} \setminus \{o_i\}$ (Plackett 1975). This probability is estimated by assigning a weight $w_i \in \mathcal{W}$ for each object o_i. The higher the value of the weight, the more preferred the object. The relationship between the probability of a specific object $o_l \in \mathcal{O}$ coming in first place ($r_l = 1$) and the weights $\mathcal{W}$ is defined as follows:

$$\mathbb{P}[r_l = 1|\mathcal{W}] = \frac{w_l}{\sum_{i=1}^{n} w_i} \tag{1}$$

The Plackett-Luce model considers each ranking order R_j as a sequence of decisions made using Equation (1). A formulation of the likelihood of this model can be found in Annex A.6. In this method, the weights $\mathcal{W}$ can be interpreted as the quality estimates of the objects. In order to follow the convention that lower latent quality estimates represent higher scores, we will transform them as follows:

$$\theta_i^{pl-lu} = (1 - w_i)$$

To estimate the weight parameters, we fit a Plackett-Luce model using an `R` implementation by Turner et al. (2020).

The Plackett-Luce model provides an estimate of the latent quality $\hat{\theta}_i^{pl-luce}$ of each object from the ranking data. Figure 8 visualises the latent quality estimates for the ten best objects according to this model. This model would accept the three objects; x25, x19, and x21. The difference in latent quality between the third place and the fourth place object is very low. The latent quality estimates are provided in Table 5.

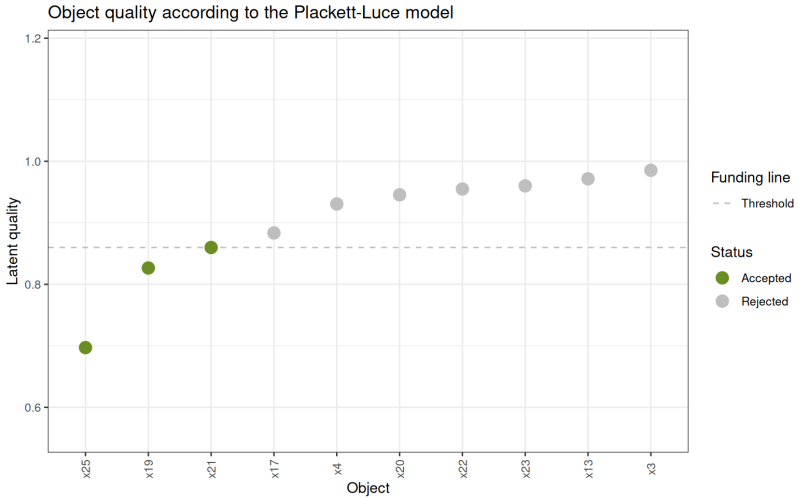


Figure 8: Latent quality estimates of the ten best objects according to the Plackett-Luce model (lower values represent higher object qualities).

Grant ID	x25	x19	x21	x17	x4	x20	x22	x23	x13	x3, x16
$\hat{\theta}_i^{pl\text{-}lu}$	0.697	0.827	0.860	0.884	0.931	0.946	0.956	0.960	0.971	0.985

Table 5: Latent quality estimates of the 10 best objects according to the Plackett-Luce model sorted from best to worst.

Mallows

The Mallows model is a distance-based ranking method which considers an underlying ranking order $R^* \in \mathcal{L}(\mathcal{O})$ and assumes the assessors are imperfect judges trying to reproduce this ranking (Mallows 1957). The probability of observing a specific ranking order $R_j \in \mathcal{L}(\mathcal{O})$ is defined as:

$$\mathbb{P}[R_j|R^*,\eta] = \frac{\exp(-\eta \;\; d(R_j,R^*))}{\psi_{\mathcal{O}}(\eta)}$$

$$d(R_j,R^*) = \sum_{i,l=1}^{n} \mathbb{1}\{o_i \;\succ_{R_j}\; o_l \quad \text{and} \quad o_l \;\succ_{R^*}\; o_i\}$$

$$\psi_{\mathcal{O}}(\eta) = \sum_{\mathcal{L}(\mathcal{O})} \exp(-\eta \;\; d(R_j,R^*)),$$

where $\psi_{\mathcal{O}}(\eta)$ is a normalizing constant, and $d(\cdot,\,\cdot)$ is the Kendall distance (minimum sum of adjacent swaps) between two linear orders. The notation $o_i \;\succ_{R_j}\; o_l$ indicates that the object o_i is ranked higher than o_l in the observed ranking order R_j. Conversely, $o_l \;\succ_{R^*}\; o_i$ indicates that o_l is ranked higher than o_i in the true ranking order R^*. The η parameter[1] represents a measure of similarity between the objects $\mathcal{O}$. A large η implies that the distribution of the ranking data R is closely concentrated around the true underlying rank R^*, while an η of 0 implies that the distribution of the ranking data R is uniform across the linear orders $\mathcal{L}(\mathcal{O})$. Using this method, the final ordering of the objects $\mathcal{F}_{mlw}(\mathcal{O})$ is $\hat{R}^*$. To fit the Mallows model, we will use an `R` implementation by Sorensen et al. (2020).

The Mallows model provides an estimate of the underlying ranking $\hat{R}^{*\;mlw}$ of the objects from the ranking data. Figure 9 visualises the rank estimates for the ten best objects according to this model. This model would accept the objects x17, x25, and x19. The ranking estimates are provided in Table 6.

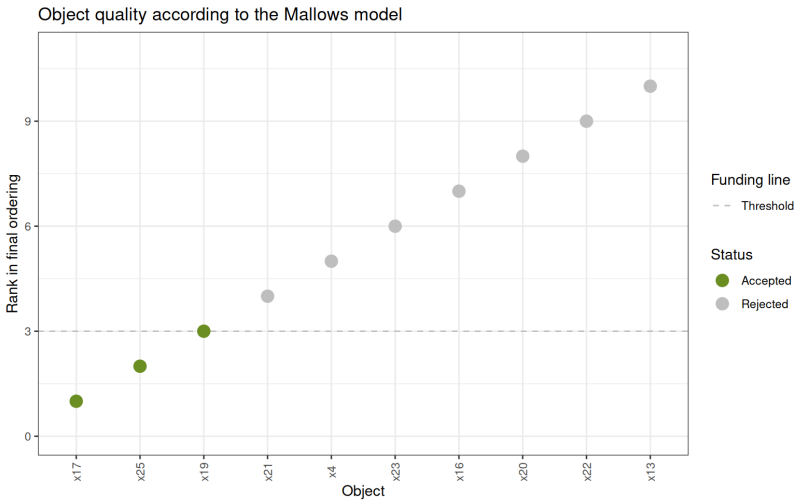


Figure 9: Underlying ranking of the ten best objects according to the Mallows model (lower values represent higher object qualities).

[1] In literature η is often denoted as θ. In this thesis, we use η, as θ is used to denote object quality.

Grant ID	x17	x25	x19	x21	x4	x23	x20	x16	x13	x22
$\hat{R}_i{}^{mlw}$	1	2	3	4	5	6	7	8	9	10

Table 6: Ranking order of the 10 best objects according to the Mallows model sorted from best to worst.

Thurstonian Model

In Thurstonian modelling, the ranking order is assumed to be determined by drawing from latent random variables z_i (as discussed in Section 1.3). The ranking orders R_j given by the assessors $j = \{1, ..., m\}$ are assumed to be determined by sorting a set of of $i = \{1, ..., n\}$ drawings of n random variable z_i in descending order. In this thesis, we use a version of the model as described in a paper by Thurstone (1927) known as a case five model to define the following:

$$z_i \sim N(\mu_i,\ \sigma^2)$$

where μ_i is a latent quality variable for object $o_i \in \mathcal{O}$ and σ^2 is the variance of the latent random variable, which can be interpreted as the noise in the perception of the quality. We add the following constraints: $\sigma^2 = 1$ in order to simplify the estimation of μ_i, and $\mu_n = 0$, in order to make the model identifiable. A formulation of the likelihood of μ using these constraints is available in Maydeu-Oliveras (2004). We then define the latent quality variable as follows:

$$\theta_i{}^{thurst} = -\mu$$

In this thesis, we fit the Thurstonian model using an `R` implementation by Revelle (2026).

The Thurstonian model provides an estimate of the latent quality $\hat{\theta}_i{}^{thurst}$ of the objects from he ranking data. Figure 10 visualises the latent quality estimates for the ten best objects according to this model. This model would accept the objects x25, x19, and x17. The latent quality estimates $\hat{\theta}_i{}^{thurst}$ are provided in Table 7.

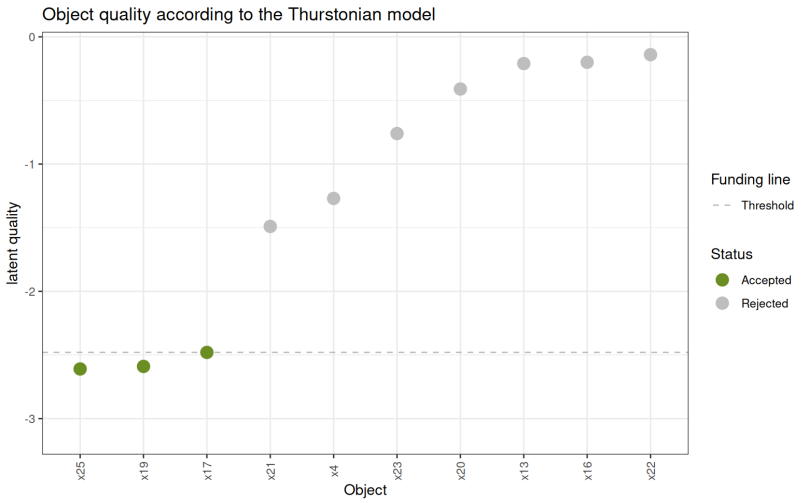


Figure 10: Latent quality of the 10 best objects according to the Thurstonian model (lower values represent higher object qualities)

Grant ID	x25	x19	x17	x21	x4	x23	x20	x13	x16	x22
$\hat{\theta}_i^{thurst}$	-2.61	-2.59	-2.48	-1.49	-1.27	-0.76	-0.41	-0.21	-0.2	-0.14

Table 7: Latent quality estimates of the 10 best objects according to the Thurstonian model sorted from best to worst.

3.3 Combined Models

Mallows-Binomial

The Mallows-Binomial model is a model based on a combination of the Mallows model for ranking and a Binomial model for scores. While the Mallows model is defined in the previous section (see Section 3.2), we have yet to define the Binomial model for scores.

The Binomial model for scores assumes that the scores given to the objects are ordinal, equally spaced and can be modelled by a Binomial distribution (Pearce and Erosheva 2022). Reformulated from the perspective of the scoring data, each object has its own Binomial distribution, and the assessors' scores given to each object are assumed to be independent observations of these distributions. The model can be formulated as follows:

$$y_i|\pi_i, l \sim Bin(\pi_i, \nu)$$
$$\mathbb{P}[y|\pi_i, h] = \binom{\nu}{y} \pi_i^x (1-\pi_i)^{\nu-x}$$

where scores have ν levels and π_i is the underlying quality of an object o_i. The expected score of an object o_i is $\nu \cdot \pi_i$.

Combining the Binomial model with the Mallows model provides the following probability distribution:

$$\mathbb{P}[R_j, S_j|\pi, \eta] = \frac{\exp(-\eta \;\; d(R_j, R^*))}{\psi_{\mathcal{O}}(\eta)} \times \prod_{i=1}^{n} \binom{\nu}{y_{ij}} \pi_i^{x_i}(1-\pi_i)^{l-x}$$
$$\pi = [\pi_1, ..., \pi_n], \quad R^* = \text{Order}(\pi), \quad \eta > 0$$

This model incorporates up to $n+1$ joint observations per assessor, with up to n scoring observations and 1 ranking observation. It jointly models the ranking and the scoring data by assuming the underlying ranking order R^* is dependent on the ordering of π. Using this method we assume the final ordering of the objects is $\hat{R}^{*\ mbin}$. In this thesis, we fit the Mallows-Binomial model using an `R` implementation by Pearce (2025). This implementation also includes a nonparametric bootstrapped uncertainty estimate, from which we provide in the results below (see Efron, B. (1979) for more information on nonparametric bootstrapping).

The Mallows-Binomial model provides an estimate of the underlying ranking $\hat{R}^{*\ mbin}$ from both the scoring data and the ranking data. This model would accept the objects x17, x25 and x19. The ranking estimates are provided in Table 8 and visualised in Figure 11.

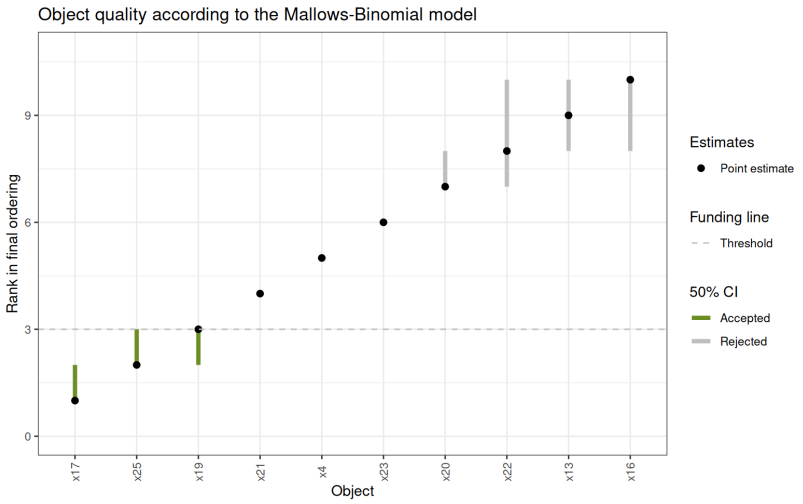


Figure 11: Rank and bootstraped uncertainty of the 10 best objects according to the Mallows-Binomial model (lower values represent higher object qualities)

Grant ID	x17	x25	x19	x21	x4	x23	x20	x22	x13	x16
$\hat{R}_i^{\ mbin}$	1	2	3	4	5	6	7	8	9	10
50% CI	1 to 2	2 to 3	2 to 3	4	5	6	7 to 8	8 to 10	8 to 10	8 to 10

Table 8: Underlying quality estimates and 50% confidence intervals (CI) of the 10 best objects according to the Mallows-Binomial model sorted from best to worst.

3.4 Comparison Between Models

We compare the six fitted models in two separate ways. First, we provide and compare the final ordering each model provides for the ten best objects. Then, in order to compare how well the models distinguish the latent quality of the objects, we perform z-scoring on the latent quality variables θ_i^k and compare the z-scores z_i^k between models. We consider that models with larger spread in the z-scores z_i^k represent models which better separate the quality between objects, and thus better distinguish the difference in quality between objects. In both the Mallows and Mallows-Binomial model a latent quality variable is not estimated by the model. We therefore perform the z-scoring on the ranks in the final ordering $\mathcal{F}_k(\mathcal{O})$ provided by each model. The formulas used to perform the z-scoring are available in Equation (2). In the context of this thesis, we are interested in the performance of the models in distinguishing the highest quality objects, we therefore perform this z-scoring uniquely on the best 10 objects according to each model.

$$z_i^k = \begin{cases} \frac{\hat{\theta}_i^k - \hat{\bar{\theta}}^k}{\hat{\sigma}_{\theta^k}}, & k = \{mean,\ hier,\ pl\text{-}lu,\ thurst\} \\ \frac{\hat{r}_i^k - \hat{\bar{r}}^k}{\hat{\sigma}_r^k}, & k = \{mlw,\ mbin\} \end{cases} \tag{2}$$

where $\hat{\bar{\theta}}^k$ and $\hat{\bar{r}}^k$ are respectively the mean score and mean rank of the ten best objects according to model k, and $\hat{\sigma}_r^k$ and $\hat{\sigma}_{\theta^k}$ are respecively the standard deviation of the scores and ranks of the ten best objects according to model k.

The results of the the comparison of final orderings can be found in Table 9 and are visualised in Figure 12. All models except the Plackett-Luce model agree that the best three proposals are x17, x25, and x19 (in no particular order). The Plackett-luce suggests instead that the best three proposals are x25, x19, and x21. The models that only take the scoring data into account suggest x17 is the best proposal, while the models that only take the ranking data into account suggest that x25 takes this position. While the models generally agree on the best three proposals, they tend to disagree about the ordering within this best three. The models also generally disagree on which objects are ranked between position 6 and 12 in the final ordering.

The results of the comparison of model performance using z-scoring can be found in Figure 13. No specific model better distinguishing object quality can be identified from this visualisation.

Position	1	2	3	4	5	6	7	8	9	10
$\mathcal{F}_{mean}(\mathcal{O})$	x17	x19	x25	x21	x4	x23	x22	x20	x6	x3
$\mathcal{F}_{hier}(\mathcal{O})$	x17	x25	x19	x21	x4	x23	x22	x20	x6	x3
$\mathcal{F}_{pl-lu}(\mathcal{O})$	x25	x19	x21	x17	x4	x20	x22	x23	x13	x3, x16
$\mathcal{F}_{mlw}(\mathcal{O})$	x17	x25	x19	x21	x4	x23	x20	x16	x13	x22
$\mathcal{F}_{thurst}(\mathcal{O})$	x25	x19	x17	x21	x4	x23	x20	x13	x16	x22
$\mathcal{F}_{mbin}(\mathcal{O})$	x17	x25	x19	x21	x4	x23	x20	x22	x13	x16

Table 9: Final ordering of the ten best objects according to all models.

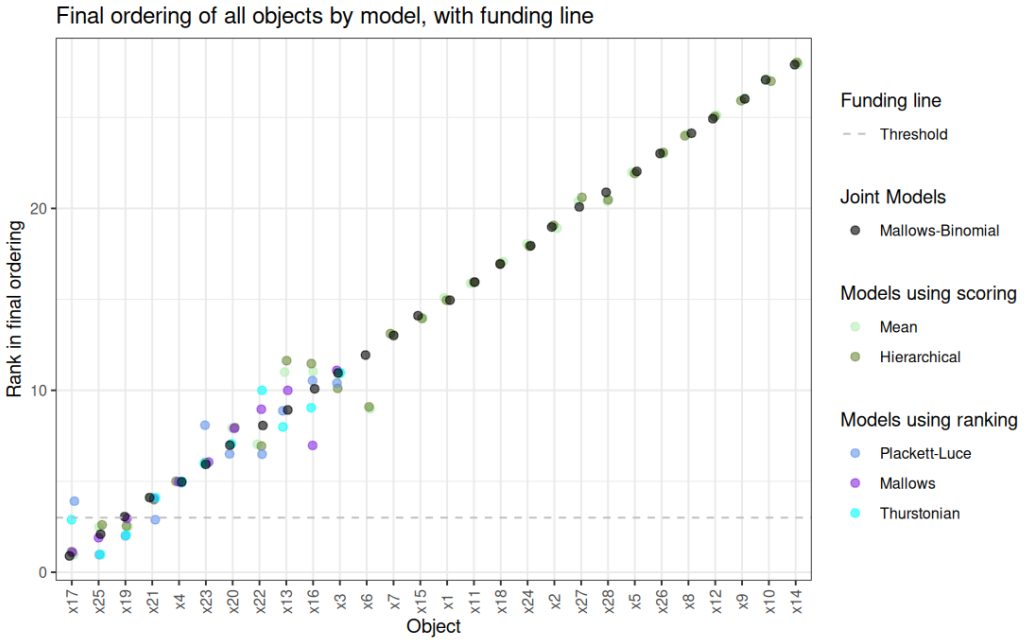


Figure 12: Plot of final ordering by model with jitter: The objects can be found on the x axis, the rank position in the final ordering can be found on the y-axis. The order of the objects was determined by the final ordering of the Mallows-Binomial model. The results from the ranking models are visualised in shades of blue, from the scoring model in shades of green, and from the joint model in black. The models solely taking the ranking data into account only provide a final ordering on the objects that have obtained at minimum one rank (11 first objects).

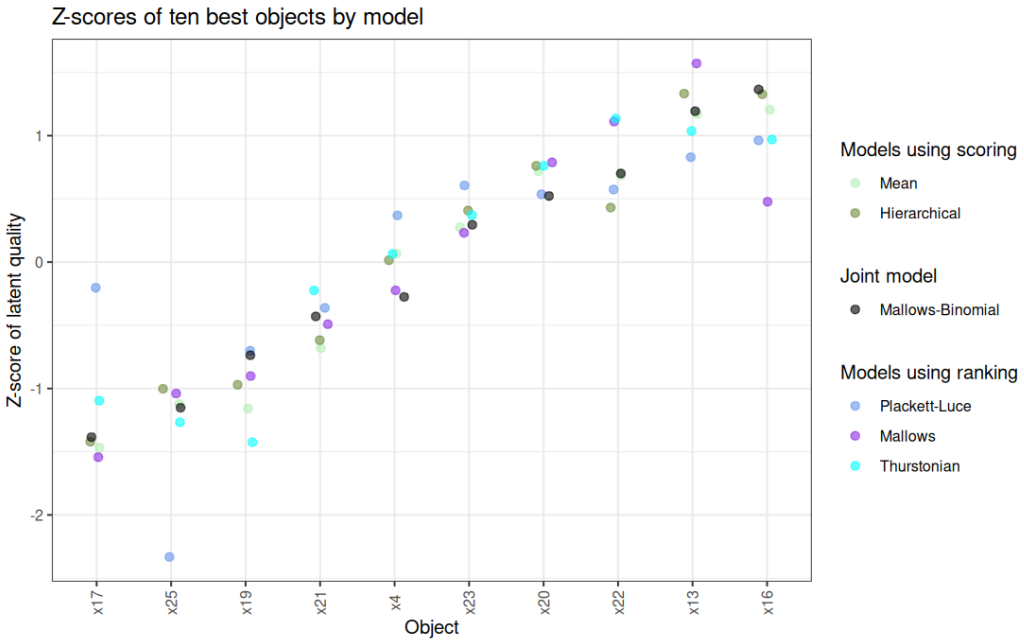


Figure 13: Plot of z-scores of the 10 best objects according to the Mallows-binomial model: The objects can be found on the x axis, the z-score of can be found on the y axis. Jitter in both the x and y direction was added for readibality. The results from the ranking models are visualised in shades of blue, from the scoring model in shades of green, and from the joint model in black.

4 Discussion

In this thesis, we provide accessible definitions of ranking data and scoring data. We use these definitions and present six different models, both frequentist and Bayesian, that can be used to analyse such data. These models are explained in great detail, with both mathematically formal and plain text definitions. We fit these models to a real-world dataset and provide their results, which we compare and discuss. We also provide reproducible code for all models, which can be used to analyse further data of the same type. All of this information provides a solid foundation for further research in analysing ranking and scoring data in the context of grant proposal quality assessment.

4.1 A Comparison of Models

Of the six models analysed, none provides the same final ordering of the objects on this dataset. Some of this variability in the results is expected; the models solely analysing the scoring data and the models solely analysing the ranking data are fitted on two different datasets, which, by the data collection procedure, do not necessarily agree with each other. However, even within models modelling the same datasets, the models still do not agree.

Within the models solely analysing scoring data, the mean score model and the hierarchical model for scores have one difference in their final orderings: the second and third position are exchanged. Upon further inspection of the data, this is due to a very small difference in the estimated proposal quality (less than 1% of the latent qualities' estimated value), which means these two proposals can be considered equivalent in quality. The reason these model provide so similar results is because they both attempt to estimate the latent quality estimate of objects using a formulation of their mean score. However, the hierarchical model takes more factors into account, such as assessor strictness. Therefore, despite, providing the same results on this dataset, we recommend using the hierarchical model for scores over the mean score model.

On the other hand, the models solely analysing ranking data result in many different final orderings. This is due to the models being based on different principles. The Plackett-Luce model estimates the probability of an object being in the first position. Therefore, it strongly favours objects which consistently obtain middling to high ranks, and disadvantages objects which polarise the assessors and obtain some very high ranks and some very low ranks. The simplified Thurstonian model has the same property as the Plackett-Luce model, although slightly less pronounced. The Mallows model works differently; it attempts to find a consensus within the ranking data. This provides an estimate that, with our ranking data, is similar to the results provided by our scoring data. The reason why these estimates are so similar remains to be investigated. These different properties show that depending on which model the user chooses to analyse their data, they are implicitly favoring some types of objects over others.

The Mallows-Binomial model joins scoring and ranking data to provide the final ordering. Using both these sources of information, the latent quality estimates of the Mallows-Binomial model seem to provide more precise estimates than the hierarchical model.

Indeed, the confidence interval bands of the Mallows-Binomial model are smaller than the credible interval bands of the Bayesian hierarchical model. While these two intervals cannot be compared one to one, looking at the results in Figure 7 and Figure 11, it seems that the Mallows-Binomial better separates the quality of the proposals than the Bayesian hierarchical model. However, despite this advantage, the Mallows-Binomial model still has some limitations. Due to its definition (Pearce and Erosheva 2022), the ranking data only contributes very little to the model in comparison to the scoring data. Indeed, the contribution to the likelihood per assessor is up to n observations for the scoring data, but only one observation for the ranking data. In total, in our dataset, this represents $n * m = 10 * 28 = 280$ scoring observations, and only $m = 10$ observations for the ranking data. By analyzing data with this model, a user implicitly decides that the scoring data is of higher importance than the ranking data.

Throughout these analyses, we show that funding agencies, when choosing a model to analyse scoring and ranking data, are directly making a decision to advandage or disadvantage some types of proposals. Any decision affecting the assessment of proposal quality should be disclosed (Harman 1998). Therefore, the choice of model, when done blindly and *post hoc*, can be considered unfair. We suggest that funding agencies should not only disclose the evaluation criteria in a call for proposal, but also disclose their data analysis procedure in order to provide a transparent and fair grant allocation.

4.2 Bayesian Model Fitting

In the description of the Bayesian hierarchical model for scores, we explore two frameworks for posterior distribution estimation: Markov-chain Monte-Carlo (MCMC) using the `rjags` package, and Integrated Nested Laplace Approximation (INLA) using the `inla` package. We provide a translation of the `rjags` code to fit the hierarchical model for scores provided in Heyard et al. (2022) into the INLA framework and show that both approaches provide very similar results. Despite the similar results, there are three important distinctions between both frameworks. First of all, MCMC, for a correctly specified model, can be shown to converge towards the true posterior distribution of a model (Hanada and Matsuura 2022), INLA on the other hand, can only provide an approximation of this distribution (Rue et al. 2009). Secondly, MCMC can be used to fit any solvable Bayesian model (Hanada and Matsuura 2022), while INLA is only restricted to latent Gaussian models, a model class which is equivalent to generalized additive models with splines (Rue et al. 2009). Finally, while MCMC model fitting is a very involved process, the use of INLA is more accessible. Indeed, MCMC model fitting requires knowledge of package specific syntax, and a significant amount of work and knowledge is required to perform convergence diagnostics (Roy 2020). INLA uses syntax very close to base `R`, and by design, does not require extensive convergence diagnoses. In our case, the Bayesian hierarchical model for scores can be formulated in the INLA framework; we therefore recommend its use over MCMC.

4.3 Paths Towards Joint Modelling

The ranking data and scoring data, while both attempting to assess the quality of objects, provide different information about them. The scoring data provides a general assessment of proposal quality on a given scale ("is this object of high or low quality?"), while the ranking data provides a direct comparative assessment of the proposal quality ("is this object of better quality than that other object?"). These two types of information should, in theory, complement each other, as the ranking information may enable us to distinguish small differences in quality between two object with very similar scores. We propose three different methods to jointly model scores and ranks.

We present in this thesis how the Mallows-Binomial model already combines scoring and ranking data (Pearce and Erosheva 2022). However, we also show that this model, while incorporating both data types, lends more importance to scoring data than ranking data. In order to create a model that considers both sources of information as equal, one could consider modifying the likelihood function of the Mallows-Binomial model to a weighted likelihood (Markatou et al. 1998). The weight of the ranking observations could then be increased and the weights of the scoring observations decreased. This would jointly model ranking and scoring data, without favouring any data type.

A second approach to joint modelling would be to create a hierarchical model in INLA, containing both a scoring and a ranking component with correlated random effects. A framework to integrate this in `R` already exists: the `inlajoint` package by Rustand et al. (2024). However, INLA can only fit models which are latent Gaussian. An implementation in this framework would require either a latent Gaussian model for ranking data, which to our knowledge does not exist, or a transformation of the ranking data into a scale which could then be used in a latent Gaussian model. This transformation could consist in using ranks as scores, which is a polarising subject in literature (Stevens 1946; Mosteller 1977; Velleman and Wilkinson 1993), or could consist in directly using the results of a Plackett-Luce, or Thurstonian model as target variable in a Gaussian model. Using one of these options would enable us to jointly model ranking and scoring data, albeit with an additional transformation step. During the course of this thesis, we created an early-stage implementation of such a model. Due to time constraints, we did not manage to refine the model enough to add it to the main part of this report. However, we still make this preliminary work available as a thesis supplement.

A third approach to joint modelling is to use the ranking data to create priors for latent quality variables, which can then be used in the Bayesian hierarchical model for scores. The priors could be created from a generalised normal Thurstonian model, as defined in Annex A.6. Unfortunately, no closed form formulation of the likelihood of of the generalised normal Thurstonian model exists, so an MCMC simulation would be necessary to estimate these distributions. A Gibbs sampling algorithm to estimate such distributions using a Thurstonian model on ranking data is already available in Yao and Böckenholft (1999). The MCMC algorithm and the Bayesian hierarchical model for scores could thus be run sequentially in order to obtain a joint model.

4.4 Limitations

This work has several limitations. The model comparisons were only performed on a single real-world dataset containing complete scoring data and partial ranking data. It is possible that the results we report and the findings we discuss are heavily influenced by the properties of these specific data and are not generalisable. Moreover, the fact that the ranking data is only partial, greatly increases the complexity of joint modelling. In this thesis, we do not discuss methods to deal with this incompleteness. However, we believe that these incomplete ranking data reflect real-world scenarios, where assessors' times are limited, and this should be taken into account in further joint modelling of ranks and scores.

In this thesis, we only describe a selection of models relevant for scoring and ranking data. This selection is neither exhaustive nor was it systematically created. Moreover, assessing the preciseness and statistical properties of the estimates provided by such models is a complicated task; different models favour different objects traits, and the decision of which traits are important depends on the real-world application of the model. In order to compare models effectively, the specific real-world objective should be defined before the analysis. Many different metrics may be suited to assess model quality, and many other models may be better suited for analysing scoring and ranking data than the ones we use in this thesis.

In the model fitting, we also only select single, specific packages for each model. While we do check whether the results for each model are plausible, we do not check whether the full back-end model implementations are correct. Moreover, in our Bayesian model, we discuss `rjags` and `inla` implementations at length, but other Bayesian posterior distribution estimation software also exist. These software implementations may be better suited for fitting the models discussed in the thesis than the ones we used.

4.5 Conclusion & Outlook

We performed the preliminary work necessary for understanding grant proposal panel data containing both scoring and ranking components. We describe six models, and find that, depending on the model selected, certain object traits may be advantaged or disadvantaged in the assessment of quality. We highlight the need for a deliberate model selection, and clear, transparent communication already during the call for proposal. Subsequently, we provide paths to potential new models that have yet to be used in this context.

We believe that in the case of grant proposal quality assessment, a "one-size fits all" approach to modelling is not the solution. In the future, we hope it will be possible to provide robust and tailored recommendations of models which fit the exact need, and reflect the exact intentions of funders. In order to obtain these robust recommendations, a thorough analysis of models, on multiple real-world and simulation datasets is still necessary. This thesis is the first step towards this path, but much work remains to be done.

A Annex

A.1 Bayesian Estimations

In frequentist models, we define fixed parameters of interest, and then subsequently define methods to find a most likely value for these parameters using data (usually by using maximum likelihood estimation) (Held and Sabanés Bové 2020). In Bayesian modelling, we also define parameters of interest. However, we assume that these parameters are not fixed, but follow distributions themselves. The focus is shifted from determining the most likely value for each parameter towards determining the most likely distribution for each parameter (Held and Sabanés Bové 2020). In order to determine these distributions we use Bayes' theorem to obtain:

$$f(\boldsymbol{\beta}|D) \propto f(D|\boldsymbol{\beta}) * f(\boldsymbol{\beta}) \tag{3}$$

where $\boldsymbol{\beta} = \{\beta_1, ..., \beta_q\}$ are the parameters of interest and D are the data used (see Annex A.2 for derivation). We denote $f(\boldsymbol{\beta}|D)$ the posterior distribution; the distribution of the parameters after having taken the data D into account, and $f(\boldsymbol{\beta})$ the prior distribution; our belief about the distribution of the parameters before taking the data into account. The function $f(D|\boldsymbol{\beta})$ is the likelihood of observing the data given the parameters of interest.

In the hierarchical model for scores (see Section 3), there exists no closed form solution to the operation $f(D|\boldsymbol{\beta}) * f(\boldsymbol{\beta})$. In such cases, in order to obtain results, we use simulations such as Markov chain Monte Carlo (Mcmc), or numerical approximation methods such as the Integrated Nested Laplace Approximations (Inla).

A.2 Bayes' Theorem

Baye's theorem is defined as follows:

$$\mathbb{P}[A|B] = \frac{\mathbb{P}[B|A] * \mathbb{P}[A]}{\mathbb{P}[B]}$$

In order to use it in modelling we reformulate it with A as the parameters of interest and B the data we have obtained. We define parameters of interest $\boldsymbol{\beta} = \{\beta_1, ..., \beta_s\}$ and data D to obtain:

$$f(\boldsymbol{\beta}|D) = \frac{f(D|\boldsymbol{\beta}) * f(\boldsymbol{\beta})}{f(D)}$$

where $f(\boldsymbol{\beta}|D)$ is the posterior distribution of $\boldsymbol{\beta}$, $f(\boldsymbol{\beta})$ the prior distribution of $\boldsymbol{\beta}$, $f(D|\boldsymbol{\beta})$ the likelihood function of our model and $f(D)$ a normalizing constant. $f(D)$ is normally estimated through marginalization:

$$f(D) = \int_{-\infty}^{\infty} f(\boldsymbol{\beta}, D) d\boldsymbol{\beta} = \int_{-\infty}^{\infty} f(D|\boldsymbol{\beta}) f(\boldsymbol{\beta}) d\boldsymbol{\beta}$$

This formulation of $f(D)$ in most cases does not have a closed form solution (unless the prior is conjugate to the likelihood). To avoid having to estimate it methods to estimate posterior distributions requiring only a proportionality constraint are used. We thus use:

$$f(\boldsymbol{\beta}|D) \propto f(D|\boldsymbol{\beta}) * f(\boldsymbol{\beta})$$

A.3 Markov Chain Monte Carlo

Markov Chain Monte Carlo (MCMC) methods are based on two concepts, Monte Carlo simulations, and Markov chains. This section will first introduce these concepts individually, and then explain how they can be used in combination to estimate a posterior distribution.

Monte Carlo

Monte Carlo simulations are a class of numerical methods that generate a series of random independent samples in order to estimate the solution to complex mathematical problems. These simulations are guaranteed, with a long enough simulation, to converge towards the analytical solution of the problem (Metropolis and Ulam 1949). Converging is defined as progressively approaching the solution of a problem as the number of simulated iterations increases (convergence is furthered discussed later in this annex). In statistics, these simulations are often used to estimate integrals of density or probability mass functions, as well as generating posterior distributions. Naïve Monte Carlo simulations work well on simple problems, but are heavily limited by the dimensionality of the problem they are solving. As dimensionality increases, the number of iterations required for convergence increase exponentially.

Markov Chain

To improve the computational performance of Monte-Carlo simulations, we can use Markov-chains. Markov chains are series of numbers $x^{(1)}, x^{(2)}, ..., x^{(q)}$ in which any step $x^{(q)}$ depends, and only depends on the previous step $x^{(q-1)}$. By using Markov-chains in our Monte Carlo simulation, we no longer use random independent samples to generate solutions to our problems. This dependant sampling is used to do importance sampling, where important domains of the solution space are sampled more than the rest of the space. This drastically reduces computation time (Metropolis et al. 1953). In order to guarantee convergence of the MCMC towards the true solution despite the dependent samples, three constraints are needed in the definition of the Markov Chain: Irreducibility, Aperiodicity and Detailed balance (Hanada and Matsuura 2022).

Definitions: Irreducibility, Aperiodicity and Detailed Balance

Markov chain: A series of numbers $x^{(1)}, x^{(2)}, ..., x^{(q)}$ in which any step $x^{(q)}$ depends, and only depends on the previous step $x^{(q-1)}$.
Irreducibility: A Markov chain is said to be irreducible if there exists a transition between any pair x and x' in a finite number of steps s.
Aperiodicity: A Markov chain is said to be aperiodic, if the greatest common divisor of all the possible number of steps $\mathbf{s} = \{s_1, s_2, ..., s_q\}$ between x and itself is 1.
Detailed balance The transition probability $\mathbb{P}[T(x \to x')]$ between any x and x' satisfy the following property:

$$\mathbb{P}[x] * \mathbb{P}[T(x \to x')] = \mathbb{P}[x'] * \mathbb{P}[T(x' \to x)]$$

Metropolis-Hasting Algorithm

Markov-chains and Monte Carlo simulations can be combined into the Metropolis-Hasting algorithm (Hastings 1970). This algorithm is the basis of the most used MCMC algorithms such as Gibbs sampling and the Hybrid Monte Carlo (HMC) algorithm. In this section, we will first introduce the Metropolis algorithm and extend it to the Metropolis-Hasting algorithm.

The Metropolis algorithm draws samples from a probability distribution $\mathbb{P}[x]$, provided that a function $f(x)$ that proportional to its distribution is available. The Markov chain in the Metropolis algorithm is generated as follows.

Definition: Metropolis algorithm

1. Initiate the Markov chain: Choose a value for $x^{(1)}$, either randomly, or using prior knowledge about the target distribution.
2. Propose a candidate $x^{(q+1)}$: Choose a real number Δx randomly and propose $x^{(q+1)} = x^{(q)} + \Delta x$. The random generation process of Δx must be symmetric around 0 in order to satisfy the irreducibility, aperiodicity and detailed balance conditions of the Markov-chain.
3. Perform the Metropolis test: The candidate $x^{(q+1)}$ is accepted with a probability:

$$\min(1, \frac{f(x^{(q+1)})}{f(x^{(q)})})$$

If the candidate is not accepted a new candidate is proposed using step two of this algorithm.
4. Repeat until convergence: Repeat from step two until the distribution of the samples generated by the Markov-chain converges.

The chain created by this algorithm satisfies irreducibility, aperiodicity and the detailed balance of the Markov chain, and the samples generated are therefore guaranteed to converge towards the target distribution, within a finite amount of iterations (Hanada and Matsuura 2022).

Metropolis-Hasting algorithms extend this algorithm by relaxing the constraint of symmetry around 0 in generating new candidates. Instead, we allow candidates to be generated using any arbitrary function, as long as the series generated is still an irreducible and aperiodic Markov chain. To still maintain the detailed balanced conditions necessary for the chain to converge the acceptance probability is modified (Hastings 1970). This provides the following algorithm:

Definition: Metropolis-Hasting algorithm

1. Initiate the Markov chain: Choose a value for $x^{(1)}$, either randomly, or using prior knowledge about the target distribution.

2. Propose a candidate value: Choose a new candidate value for the chain using an arbitrary function$f(x^{(q+1)}|x^{(q)})$.
3. Perform the Metropolis test: The candidate $x^{(q+1)}$ is accepted with a probability:

$$\min(1, \frac{f(x^{(q+1)}) * f(x^{(q)}|x^{(q+1)})}{f(x^{(q)}) * f(x^{(q+1)}|x^{(q)}))}$$

If the candidate is not accepted a new candidate is proposed using step two of this algorithm.
4. Repeat until convergence: Repeat from step two until the distribution of the samples generated by the Markov-chain is converges

The Gibbs sampling algorithm and the HMC algorithm are special cases of the Metropolis-Hasting algorithm with precisely specified candidate value functions $f(x^{(q+1)}|x^{(q)})$.

Gibbs Sampling

Gibbs sampling is a Metropolis-Hasting algorithm used in order to estimate multivariate target distributions. Let us consider a target distribution $\mathbb{P}[\mathbf{x} = x_1, ..., x_r]$ with a proportional distribution $f(\mathbf{x})$ (Geman and Geman 1984). In Gibbs sampling, the candidate values $\mathbf{x}^{(q+1)}$ are generated by using sequential conditional distributions:

$$x_a^{(q+1)} \sim f(x_a|x_1^{(q+1)}, ..., x_{a-1}^{(q+1)}, x_{a+1}^{(q)}, ..., x_r^{(q)}) \quad (4)$$
$$\text{For all } x_r \in \mathbf{x}$$

This model is sequential and updated values at stage $(q+1)$ and values at stage (q) are both incorporated together in order to obtain new candidate values. Moreover, the distribution of $x_a^{(q+1)}$ depends on all other variable x_{-a} but not on $x_a^{(q)}$ itself. Using such a generation process for $\mathbf{x}^{(q+1)}$ provides the following Metropolis test in the Gibbs sampling algorithm:

$$\mathbb{P}[\text{acceptance}] = \frac{f(x_r^{(q+1)}) * f(x_r^{(q)})|x_r^{(q+1)})}{f(x_r^{(q)}) * f(x_r^{(q+1)}|x_r^{(q)})}$$

and by definition:

$$x_r^{(q+1)} \perp\!\!\!\perp x_r^{(q)}$$

thus,

$$\mathbb{P}[\text{acceptance}] = \frac{f(x_r^{(q+1)}) * f(x_r^{(q)})}{f(x_r^{(q)} * f(x_r^{(q+1)})} = 1$$

Thus, the Metroopolis test is not needed in order to obtain the distribution $\mathbb{P}[\mathbf{x}]$. Gibbs sampling was originally defined by Stuart and Donald Geman (1984). However, we use here a simpler definition as formulated in Hanada and Matsuura (2022):

Definition: Gibbs sampling algorithm

1. Initiate the Markov chain: Choose a value for $\mathbf{x^{(1)}} = \{x_1^{(1)}, ...x_q^1\}$, either randomly, or using prior knowledge about the target distribution.
2. Propose a new value for each $\mathbf{x}^{(q+1)}$: Using the procedure described in equation (4)
3. Repeat until convergence: Repeat from step two until the distribution of the samples generated by the Markov-chain is converges.

Posterior Distribution Estimation using Markov Chain Monte Carlo

All the definitions in this section were described as estimating a distribution $\mathbb{P}[x]$ given an arbitrary proportional function $f(x)$. To obtain the posterior distribution as described in Equation (3). We reformulate the MCMC algorithm as follows:

$$\mathbb{P}[x] \propto f(x)$$
$$f(\beta|D) \propto f(D|\beta) \cdot f(\beta)$$

Thus, we can sample from a likelihood $f(D|\beta)$ and a prior distribution $f(\beta)$, in order to estimate a posterior distribution $f(\beta|D)$.

Trace Plots and Autocorrelation

Trace plots are often used to assess the convergence and quality of a MCMC simulation. Trace plots are plots that shows the progression of the sampled parameter (y-axis) across the Markov chain's iterations (on the x-axis). On such plots, we can identify if models converge and the number of iterations taken in order to converge (see Figure 14, left). If the parameter is stably oscillating around a parameter value, we consider the model as converged.

As can be seen in Figure 14, left, the first iterations of an MCMC simulation are heavily correlated with the Markov chain's starting value. Moreover, even when a Markov-chain converges, the samples it generates are autocorrelated. In order to limit the effect of these correlations on the solution provided by MCMC simulations, we use burn-in and thinning (Hanada and Matsuura 2022). Burn-in consists in discarding the samples in the Markov-chain that are deemed correlated with the starting value, to determine the number of samples to discard, trace plots are often used. Thinning is used in an attempt to eliminate the correlation between samples after the burn-in period. It involves only taking a fraction of the observation produced by the Markov chain (for example 1 in 10 or 1 in 20 samples in the chain) in order to estimate the distribution of the parameter of interest. If this skipping jump is longer than the autocorrelation structure, we obtain independent samples. Multiple methods exist to determine the correct thinning size. A complete description of these methods is beyond the scope of this thesis.

Convergence and Diagnostics

In Section A.3 we state that given a long enough chain, a MCMC simulation will always converge towards the true analytical solution of a problem. This, while true, is a misleading statement. Indeed, as simulations are limited by computing capabilities, it is not always possible to run a chain until it converges towards the solution of a problem. Moreover, if a model is misspecified there may be no stationary solution in the first place.

There exist multiple methods to check if a model has converged. A first check to see if a simulation has converged, is by running it multiple times with different starting values (in MCMC this is denoted as multiple chains). A trace plot to visualise all chains (see Figure 14, right) can then be used to check if chains converge to the same solution. However, this is only a visual test, many other statistical diagnostics for convergence exist. A complete description of these methods is beyond the scope of this thesis. However, we will describe the intuition behind three methods we use in the assessment of convergence of the Bayesian hierarchical model for scores.

Gelman and Rubin (1992) propose a $\hat{R}$ measure (also called potential scale reduction measure) to assess the convergence of MCMC simulations. It consists in calculating two measures of variance in the chains generated by a simulation. One measure calculates the variance of the samples generated within each chain, while the other measures the variance of the samples generated by all chains together. If these two measures differ, it is assumed that the individual chains have not yet had enough iterations to range over the stationary distributions. The $\hat{R}$ measure is a division of both these measures, an $\hat{R}$ close to one therefore signifies good convergence. A multivariate version of the $\hat{R}$ measure was later developped by Brooks and Gelman (1998).

The Heidelberger and Welch diagnostic (Heidelberger and Welch 1983) is a two-stage test assessing the convergence of an MCMC chain. First, it compares the cumulative distribution of the samples generated by a markov chain to the cumulative distribution of a Brownian bridge (which is assumed to be what a stationary MCMC chain should sample like). The Cram'er-von-Misis test is used to test the similarity between the two distributions. In a second change, Heidelberger and Welch propose a precision test in order to assess if the model estimates are precise enough.

The last diagnostic we use in this thesis is the Raftery and Lewis diagnostic (Raftery and Lewis 1970). However, it is not used to estimate convergence, but instead used to estimate the length (number of iterations) of an MCMC simulation necessary in order to estimate a quantile q at an accuracy of r with a probability p. It is based on an indicator function which counts how many samples generated in a MCMC simulation fall within an accuracy r of q.

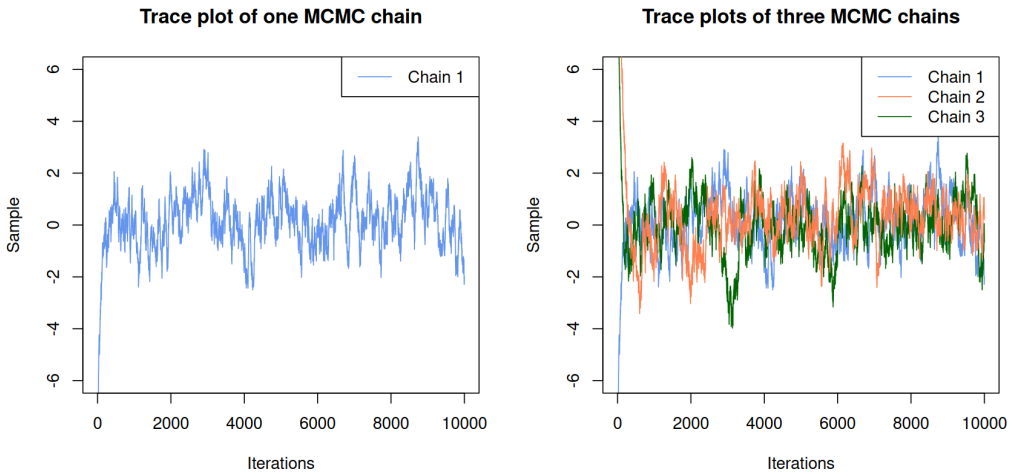


Figure 14: Trace plots of an MCMC simulation estimating a normal distribution with a mean of 0 and standard deviation of 1. **On the left:** Trace of a single chain with visible burn-in period and autocorrelation structures. **On the right:** Trace of three chains converging towards the same solution.

A.4 Integrated Nested Laplace Approximation

The Integrated Nested Laplace Approximation (INLA) is a method to approximate posterior distributions based on the Laplace approximation. It uses the approximation multiple times, in nested iterations, to obtain the marginal posterior distribution of the parameters of a latent Gaussian model. A thorough description and motivation of the INLA method is available in the original paper by Rue et al. (2009). An extension of this method enabling joint modelling is available in Rustand et al. (2024, 2026). While describing the full construct of INLA is beyond the scope of this thesis, we offer here a short introduction to the Laplace approximation.

Laplace Approximation

The Laplace approximation can be used to numerically approximate a function $f(x)$ around it's mode, using a normal distribution (Held and Sabanés Bové 2020). We take:

$$h(x) = \log(f(x))$$

and approximate it around the mode x_0 of $f(x)$ using a 2nd order Taylor expansion:

$$\begin{aligned} h(x) &\approx h(x_0) + h'(x_0)(x - x_0) + \frac{h''(x_0)(x - x_0)^2}{2} \\ &\approx h(x_0) + \frac{h''(x_0)(x - x_0)^2}{2} \end{aligned}$$

which then provides:

$$\begin{aligned} f(x) &= e^{h(x)} \\ &\approx e^{h(x_0)+\frac{h''(x_0)(x-x_0)^2}{2}} \\ &\approx f(x_0) * \underbrace{e^{-\frac{1}{2}\frac{(x-x_0)^2}{-h''(x_0)^{-1}}}}_{\propto N(\mu=x_0,\ \sigma^2=-h''(x_0)^{-1})} \end{aligned}$$

This can used to estimate the integral of a function $f(x)$ which would otherwise be hard to integrate.

A.5 Generalisation of the Normally Distributed Thurstonian Model

We modify the normally distributed Thurstonian model defined in Yao and Böckenholt (1999) to define the following Thurstonian model:

$$z_i \sim N({\theta_i}^{thurst},\ \sigma_i^2)$$

where ${\theta_i}^{thurst}$ is the latent quality variable for object $o_i \in \mathcal{O}$ and σ_i^2 is the variance of the latent random variable, which can be interpreted as the noise in the perception of the quality. This can be formulated in a general setting:

$$\mathbf{Z_j} \sim N(\boldsymbol{\theta},\ \boldsymbol{\Sigma})$$

The vector of all scores $\mathbf{Z_j} = \{z_1, ..., z_n\}$ drawn by an assessor j can be defined as an n-variate normal distribution with mean $\boldsymbol{\theta} = \{{\theta_1}^{thurst}, ..., {\theta_n}^{thurst}\}$ and covariance matrix $\boldsymbol{\Sigma} = diag(\sigma_1, ..., \sigma_n)$.

There exists no closed-form analytical solution to estimate the values of $\boldsymbol{\theta}$ and $\boldsymbol{\Sigma}$ from ranking data, an open form solution can be found in Annex A.6. However, this estimation procedure is very computationally intensive due to the high amount of integration required to obtain results. To solve this issue, parameter estimation can be done using Gibbs sampling (Yao and Böckenholt 1999; Johnson and Kuhn 2013). To our knowledge, no implementation of this algorithm in `R` is currently available online.

A.6 Likelihoods

Plackett-Luce

Silverberg (1980) states that the Plackett-Luce ranking process can be regarded as a series of drawing from a vase. We consider a vase containing an infinite amount of balls of n colours, each colour associated with an object $o_i \in \mathcal{O}$. The frequency of each specific ball colour in the vase is proportional to the weights W assigned to O. We then draw in a series of phases from the vase. In the first phase, the first colour drawn determines the object in first position of the ranking. In each subsequent phase, the first colour drawn that has not already been drawn determines the next objects in the ranking. We perform this procedure until all ball colours (all objects) are drawn.

Using the vase model representation of the Plackett-Luce model, we can state the probability of observing a specific linear order as follows:

$$\mathbb{P}[R_j|\mathcal{W}] = \frac{w_1}{\sum_{i=1}^{n} w_i} * \frac{w_2}{\sum_{i\neq 1}^{n} w_i} * ... * \frac{w_n}{w_n}$$

If we consider a set $\mathcal{O}_x$, the set of objects that have not been ranked yet at stage x of the model and o_{i_x} the object selected at stage x of the model, we can rewrite the likelihood more compactly:

$$\mathbb{P}[Rj|\mathcal{W}] = \prod_{x=1}^{n} \frac{w_{i_x}}{\sum_{i\in\mathcal{O}_x} w_i}$$

using this, we obtain the log-likelihood provided by one observation:

$$\begin{aligned} l(\mathcal{W}|R_j) = \mathbb{P}[R_j|\mathcal{W}] &= \log\Big(\prod_{x=1}^{n} \frac{w_{i_x}}{\sum_{i\in\mathcal{O}_x} w_i}\Big) = \sum_{x=1}^{n} \log\Big(\frac{w_{i_x}}{\sum_{i\in\mathcal{O}_x} w_i}\Big) \\ &= \sum_{x=1}^{n} \Big(\log(w_{i_x}) - \log(\sum_{i\in\mathcal{O}_x} w_i)\Big) \end{aligned}$$

which in turn provides the general log-likelihood function:

$$l(\mathcal{W}|R) = \sum_{j=1}^{m} \left(\sum_{x=1}^{n} \Big(\log(w_{i_x}) - \log(\sum_{i\in\mathcal{O}_x} w_i)\Big) \right)$$

Thurstonian

We first define the model as specified in Yao and Böckenholt (Yao and Böckenholt 1999):

$$\mathbf{Z_j} \sim N(\bar{\mathbf{z}} + \boldsymbol{\theta},\ \boldsymbol{\Sigma})$$

Using $\mathbf{Z}$, the probability of a specific object o_i being ranked higher than an other object o_l in a ranking order can be written as:

$$\begin{aligned} \mathbb{P}[o_i \succ o_l|\boldsymbol{\theta}, \boldsymbol{\Sigma}] = \mathbb{P}[z_i \geq z_l] &= \mathbb{P}[z_i - z_l \geq 0] \\ &= \int_0^{\infty} \phi(z_i - z_l) \\ &= \int_0^{\infty} \frac{1}{\sqrt{2\pi(\sigma_i + \sigma_l)}} \exp(-\frac{(x - (\theta_i - \theta_l))^2}{2(\sigma_i + \sigma_l)^2}) \end{aligned} \tag{5}$$

Moreover, the following property holds:

$$\{o_i \succ o_l\} \wedge \{o_l \succ o_p\} \implies \{o_i \succ o_p\} \tag{6}$$

Using equation (5, 6) we can then define the probability of observing a full ranking order $R_j = [o_1 \succ o_2 \succ ... \succ o_n] \in \mathcal{L}(\mathcal{O})$ as:

$$\begin{aligned}
\mathbb{P}[R_j|\boldsymbol{\theta}, \boldsymbol{\Sigma}] &= \mathbb{P}[z_1 \succ z_2 \ \cap \ z_2 \succ z_3 \ \cap \ ... \ \cap z_{n-1} \ \succ \ z_n] \\
&= \mathbb{P}[z_1 - z_2 \geq 0] \ \cdot \ \mathbb{P}[z_2 - z_3 \geq 0] \ \cdot \ ... \ \cdot \ \mathbb{P}[z_{n-1} - z_n \geq 0] \\
&= \int_0^\infty \phi(z_1 - z_2) \cdot \ ... \ \cdot \int_0^\infty \phi(z_{n-1} - z_n)
\end{aligned}$$

which in turn, can be inserted in the likelihood function:

$$L(\boldsymbol{\theta}, \boldsymbol{\Sigma}|R) = \prod_{j=1}^{m} \mathbb{P}[R_j|\boldsymbol{\theta}, \boldsymbol{\Sigma}]$$

and log-likelihood function

$$l(\boldsymbol{\theta}, \boldsymbol{\Sigma}|R) = \sum_{j=1}^{m} \log(\mathbb{P}[R_j|\boldsymbol{\theta}, \boldsymbol{\Sigma}])$$

A.7 Convergence Diagnostics in the Hierarchical Model for Scores

Convergence diagnostics were performed using the CODA package (Plummer et al. 2006). The posterior distribution of all parameters converged with $\hat{R} = 1$ point estimates and a multivariate potential scale reduction factors of 1. This indicates very good convergence (Brooks and Gelman 1998). The autocorrelation factors of all parameters dropped below 0.05 after a lag of 20 iterations (see Figure 15 for a selection of autocorrelation factor plots). The thinning parameter of 20 used in the fit therefore seems appropriate. No trace plot showed any sign of abnormal behavior across the four MCMC chains (see Figures 16 and 17 for a selection of trace plots). The Raftery and Lewis diagnostic (Raftery and Lewis 1970) suggests a minimum of 3746 iterations for a 95% probability of achieving an accuracy of 0.005 in the estimation of the 2.5% quantile. Our model was fitted using 80000 iteration and with a thinning parameter of 20, we therefore consider the number of iteration sufficient. We also tested for the stationarity of the posterior distributions of all parameters on all four chains using Heidelberger's convergence diagnostic (Heidelberger and Welch 1983). All parameter's estimated by the model passed the test. Due to the model passing all these tests, and visually showing satisfactory properties in the trace plots, we assume the model successfully converged.

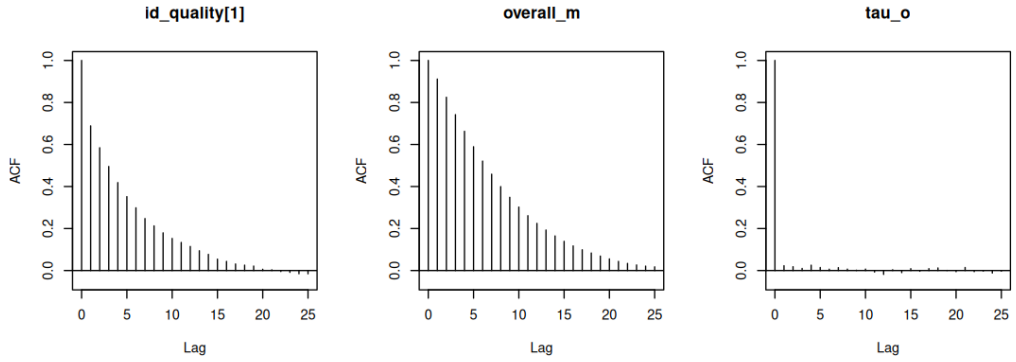


Figure 15: Autocorrelation factor of a selection of parameters estimated using the `rjags` implementation of the hierarchical model.

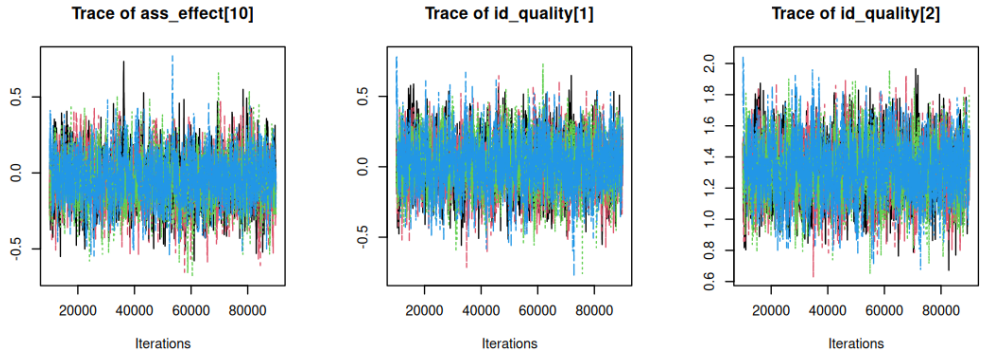


Figure 16: Trace plot of a selection of assessor and object quality parameters estimated using the `rjags` implementation of the hierarchical model.

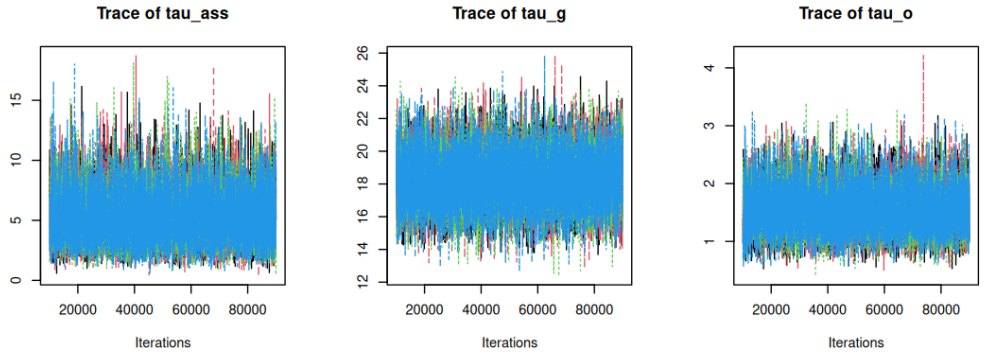


Figure 17: Trace plot of the standard deviation hyperparameters estimated using the `rjags` implementation of the hierarchical model.

A.8 Model Calls and Priors in the Hierarchical Model for Score

In this section we provide the models called in the JAGS and INLA implementations of the Bayesian hierarchical models for scores. Plots of the hyperpriors used in these models are also available in this section in Figure 18. Finally, an overview of the main properties of these hyperpriors is available in Table 10.

```
### rjags call ###################################################################

model_string <- '
  model  {

  # Likelihood -------------------------------------------------------------

    for (i in 1:n){
      score[i] ~ dnorm(mu[i], tau_g)
      mu[i] <- overall_m + id_quality[object_id[i]] + ass_effect[assessor[i]]
    }

  # Priors for id and ass effect -------------------------------------------

    for(j in 1:n_object){
      id_quality[j] ~ dnorm (0, tau_o)
    }

    for(k in 1:n_assessor){
      ass_effect[k] ~ dnorm(0, tau_ass)
    }


  # Prior for overall_m  ---------------------------------------------------

    overall_m ~ dnorm(0, 10^(-3))

  # Hyperpriors ------------------------------------------------------------

    tau_g ~ dgamma(1,0.1) # This precision is high as model very flexible

    tau_o ~ dgamma(1,1) # matched with inla prior

    tau_ass ~ dgamma(1,1) # matched with inla prior

  }
'

# Model defintion
```

```r
model <- jags.model(textConnection(model_string),
                    data = jags_dat,
                    n.chains = 4,
                    n.adapt = 10000)

# Burn in
update(model, n.iter = 10000)

# Sampling
fit <- coda.samples(model = model,
                    variable.names = c("overall_m",
                                       "id_quality",
                                       "ass_effect",
                                       "tau_g", "tau_o", "tau_ass"),
                    n.iter = 80000,
                    thin = 20)

### INLA call ##################################################################

# Setting prior for random effects
# According to INLA documentation "loggamma" is the gamma prior
prec.prior <- list(prec = list(prior = "loggamma", param = c(1, 1)))


# Model call
fit_inla <-
  inla(formula = value ~ 1 + f(grant_id, model = "iid", hyper = prec.prior) +
                 f(RandomRevID, model = "iid", hyper = prec.prior),
                 data = dat, quantiles = c(0.25,0.75),
                 # Setting intercept prior
                 control.fixed =
                    list(mean.intercept = 0, prec.intercept = 10^(-3)),
                 # Setting error term prior
                 family = "gaussian",
                 control.family = list(
                   hyper = list(
                     prec = list(prior = "loggamma", param = c(1, 0.1))))
       )
```

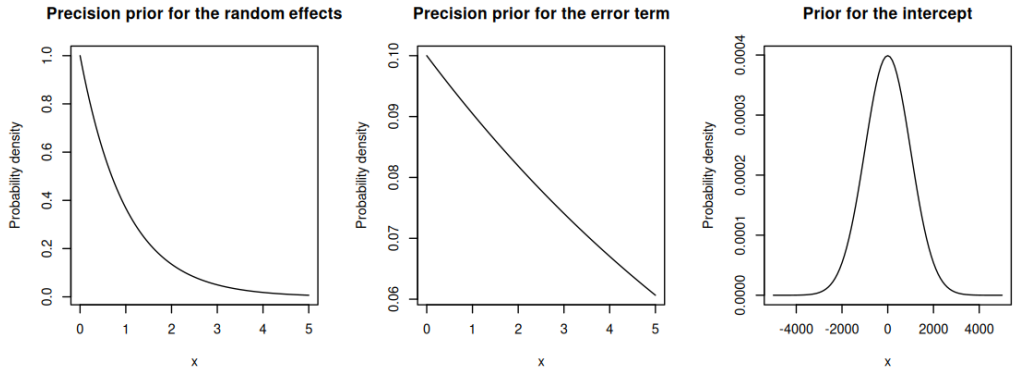


Figure 18: Visualisation of the hyperpriors used in the hierarchical model for scores. The precision priors provide plausible values for the precision of the random effects and intercepts. The intercept prior is uninformative. All three priors were selected out of conveniance, as they are straightforward to implement.

prior	25% quantile	mean	75% quantile
precision for the random effects	0.29	1	1.39
precision for the error term	2.88	10	13.86
Intercept	-674.5	0	674.5

Table 10: Description of the priors used in the hierarchical model

A.9 Code Availability

All code used in order to write this report, analyse data, and generate figures in this thesis and its supplement are available on github under the following link: https://github.com/cchipmunk/MaTh-funding-rank-score

A.10 Use of AI Declaration

I hereby declare that the work carried out in this thesis was done independently, and that no part of this project, including all code, reports and presentations was written by any kind of generative artificial intelligence tool. I also declare that no artificial intelligence tools were used for any kind of proofreading or reformulating in this project. I acknowledge that work that violates the principles set out in this declaration of authenticity may have legal and disciplinary consequences.

A.11 Session Info

```
R version 4.6.2 (2026-10-29)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.5 LTS
```

```
Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0  LAPACK version 3.12.0

locale:
 [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
 [9] LC_ADDRESS=C               LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: Europe/Zurich
tzcode source: system (glibc)

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

loaded via a namespace (and not attached):
 [1] compiler_4.6.2    fastmap_1.2.0     cli_3.6.6         tools_4.6.2
 [5] htmltools_0.5.9   otel_0.2.0        rstudioapi_0.18.0 yaml_2.3.12
 [9] rmarkdown_2.31    knitr_1.51        jsonlite_2.0.0    xfun_0.57
[13] digest_0.6.39     rlang_1.2.0       evaluate_1.0.5
```

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